



EE214 Digital Circuits Laboratory

Wadhvani Electronics Laboratory
Electrical Engineering IIT Bombay

Problem set: 2

Date: August 14, 2026

Designing MUX

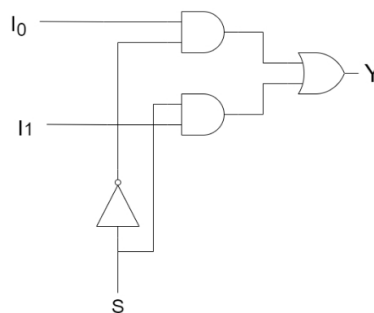
Instructions:

1. Use **structural** modelling for this experiment; i.e. instantiating components and using port map to establish the connection between those components.
2. Draw the pen-and-paper design of each circuit, with proper labels for all wires. Use the same labels in the corresponding VHDL code.
3. The pen-and-paper design of each circuit must be presented step-by-step and should include all necessary **Truth Tables, Karnaugh maps, and the resulting logic equations.**
4. Perform RTL simulation using the provided testbench and TRACEFILE to verify the design.

Problem Statement:

1. Part-A: 2x1 Mux

- (a) VHDL description: Write the VHDL description of a 2x1 multiplexer as shown in figure below.
INFO: Multiplexer is a combinational circuit that has maximum of 2^n data inputs, 'n' selection lines and single output lines.



- (b) Write Truth table and boolean expression for output Y.
- (c) Simulation: Simulate the 2x1 multiplexer using the generic testbench to confirm the correctness of your description.

NOTE: To do this, use the tracefile given below and modify the testbench given to you appropriately.

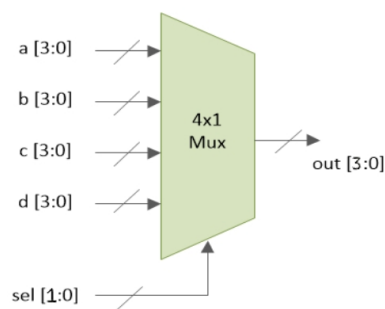
Tracefile format: ($\langle In1 \rangle \langle In0 \rangle \langle S \rangle \langle Y \rangle$ 1) [Tracefile](#)

2. Part-B: 4x1 Mux

- (a) Design: Design 4x1 Mux using only 2x1 Mux
INFO: $2^2 = 4$ data inputs, 2 select lines, 1 output line
- (b) VHDL description: Write the VHDL description of a 4x1 multiplexer designed using 2x1 Mux.
- (c) Simulation: Simulate the 4x1 multiplexer using the generic testbench to confirm the correctness of your description.
NOTE: To do this, use the given tracefile and modify the testbench given to you appropriately.
Tracefile format: ($\langle In4 \rangle \langle In3 \rangle \langle In2 \rangle \langle In1 \rangle \langle S2 \rangle \langle S1 \rangle \langle Y \rangle$ 1) [Tracefile](#)

3. Part-C: 4-bit 4x1 Mux

- (a) Design: Design 4-bit wide 4x1 Mux using only 4x1 Mux designed in part B.



- (b) VHDL description: Write the VHDL description of a 4-bit wide 4x1 multiplexer designed using 4x1 Mux.
- (c) Simulation: Simulate the 4-bit 4x1 multiplexer using the generic testbench to confirm the correctness of your description.
NOTE: To do this, use the tracefile given below and modify the testbench given to you appropriately.
Tracefile format:
($d3\ d2\ d1\ d0 \rangle \langle c3\ c2\ c1\ c0 \rangle \langle b3\ b2\ b1\ b0 \rangle \langle a3\ a2\ a1\ a0 \rangle \langle sel1\ sel0 \rangle \langle Y3\ Y2\ Y1\ Y0 \rangle$ 1111)
[Tracefile](#)